

SMART REAL-TIME AGE, GENDER, EMOTION, AND BEHAVIOR-BASED ADVERTISING SYSTEM FOR SRI LANKAN MALLS

Project ID: 25-26J-091

Final Report

BSc (Hons) degree in Information Technology
Specializing in Data Science

Department of Computer Science
Sri Lanka Institute of Information Technology
Sri Lanka

April 2026

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DECLARATION

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We declare that this is our own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, we hereby grant to Sri Lanka Institute of Information Technology, the non-exclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. We retain the right to use this content in whole or part in future works (such as articles or books).

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ABSTRACT

Retail environments increasingly require intelligent systems capable of understanding customer demographics and behavior in real time to enhance personalization and improve marketing effectiveness. This study presents a smart retail analytics and advertisement recommendation system that integrates age and gender detection, emotion recognition, and behavioral engagement analysis within a unified framework.

The proposed system utilizes computer vision and machine learning techniques to process live video streams captured through a camera. Facial features are extracted and analyzed using Convolutional Neural Network (CNN)-based models to predict demographic attributes such as age group and gender, as well as emotional states. In addition, user engagement is estimated using observable behavioral cues, including attention duration, head orientation, motion patterns, and interaction consistency. These analytical outputs are aggregated to form a comprehensive user profile that reflects both static and dynamic user characteristics.

A rule-based recommendation engine leverages this profile to generate personalized advertisements tailored to the detected user attributes and engagement levels. The system is implemented using technologies such as OpenCV for video processing, TensorFlow/Keras for deep learning model development, FastAPI for backend integration, and MongoDB for data storage and management. The architecture is designed to support real-time processing with minimal latency, enabling immediate adaptation of displayed content based on user behavior.

The performance of the system is evaluated using a combination of publicly available datasets and real-time testing scenarios to assess accuracy, responsiveness, and robustness under varying environmental conditions. Experimental results indicate that the proposed system achieves satisfactory accuracy in demographic and emotion classification while maintaining efficient real-time performance. Furthermore, the system successfully generates context-aware advertisement recommendations that align with user profiles.

The findings demonstrate the feasibility and practical applicability of intelligent vision-based systems in retail environments, highlighting their potential to enhance customer engagement, improve advertisement effectiveness, and support data-driven decision-making in modern retail ecosystems.

Keyword: Computer Vision, Emotion Recognition, Customer Analytics, Advertisement Recommendation, Retail Intelligence

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to our supervisor, Dr. Nathalie Silva, for her continuous guidance, invaluable feedback, and encouragement throughout the duration of this research project. Her support has been instrumental in shaping this study from its inception to completion. I also extend my sincere appreciation to my co-supervisor, Mrs. Shashini Kumarasinghe, for her consistent support and insightful input throughout the project. Lastly, I am deeply grateful to my teammates, friends, and family for their support, encouragement, and understanding throughout this journey.

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INTRODUCTION

In recent years, the rapid advancement of digital technologies has significantly transformed the way businesses understand and interact with customers. Online platforms leverage user data extensively to deliver highly personalized experiences, enabling targeted advertising and improved customer engagement [1] , [2]. In contrast, traditional physical retail environments still rely heavily on static advertising methods and manual observation, limiting their ability to dynamically adapt to customer preferences.

With the emergence of Artificial Intelligence (AI) and Computer Vision technologies, it has become possible to extract meaningful insights from visual data in real time [3]. These technologies enable systems to detect and interpret human attributes such as age, gender, facial expressions, and behavioral patterns. Such capabilities present a valuable opportunity to enhance customer interaction within physical retail spaces by introducing intelligent, data-driven decision-making mechanisms.

One of the key challenges faced by retail environments is the lack of real-time customer analytics. Retailers are often unable to accurately identify the demographics and emotional states of customers interacting with displays or kiosks. As a result, advertisements are typically generic and fail to capture the attention of diverse customer groups. Personalized advertisement delivery, which has proven highly effective in digital platforms, remains largely underutilized in physical environments [1].

The system is proposed as a solution to address this gap by integrating multiple AI-driven components into a unified system. The system utilizes computer vision techniques to analyze live video streams captured through a camera. It identifies customer demographic attributes such as age group and gender, detects emotional states through facial expressions, and estimates behavioral engagement based on observable cues such as attention duration and interaction patterns. These insights are then used to recommend context-aware advertisements tailored to the detected user profile.

By combining demographic analysis, emotional understanding, and behavioral engagement within a single pipeline, system aims to bring intelligent personalization into physical retail environments. This approach has the potential to improve customer experience, increase advertising effectiveness, and provide valuable insights into business decision-making [2].

In addition to improving customer engagement, the integration of intelligent analytics within retail environments also supports data-driven decision-making for businesses. Traditionally, retailers rely on sales data and manual observations to understand customer behavior, which often provides only a partial and delayed view of customer interactions. In contrast, real-time analytics systems enable immediate insight into how customers respond to products, advertisements, and in-store

experiences. This shift from reactive to proactive decision-making represents a significant advancement in retail strategy.

Another important aspect is the growing expectation of personalization among modern consumers. Customers are increasingly accustomed to personalized recommendations in digital environments, such as e-commerce platforms and social media. This expectation is gradually extending to physical retail spaces, where customers seek more engaging and relevant experience. However, the lack of intelligent infrastructure in traditional retail settings creates a gap between customer expectations and actual service delivery. Addressing this gap requires the adoption of technologies capable of understanding customer characteristics and adapting content in real time.

The application of computer vision in retail environments also introduces opportunities for non-intrusive data collection. Unlike traditional methods that rely on explicit user input or tracking mechanisms, vision-based systems can analyze customer attributes without requiring direct interaction. This allows for seamless integration into existing retail setups, such as digital signage, kiosks, and interactive displays. However, such implementations must carefully consider ethical and privacy concerns, ensuring that data is handled responsibly and that user anonymity is preserved.

Moreover, the dynamic nature of customer behavior presents additional challenges that require robust system design. Factors such as varying lighting conditions, occlusions, crowd density, and diverse demographic characteristics can affect the accuracy and reliability of visual analysis systems. Therefore, an effective solution must not only incorporate advanced machine learning models but also ensure resilience and adaptability in real-world conditions. This highlights the importance of designing systems that balance accuracy with practical usability.

The system addresses these challenges by providing an end-to-end solution that integrates multiple analytical components into a cohesive framework. Unlike standalone systems, MallVision emphasizes the interaction between different modules, enabling a more comprehensive understanding of customer behavior. For instance, combining emotional analysis with engagement metrics allows the system to distinguish between passive presence and active interest, leading to more meaningful advertisement recommendations.

Furthermore, the system is designed with real-time processing capabilities, allowing immediate adaptation of displayed content based on detected user attributes. This real-time responsiveness is critical in retail environments, where customer attention spans are limited and decision-making occurs within short time frames. By delivering relevant advertisements at the right moment, system enhances the likelihood of capturing customer interest and influencing purchasing behavior.

In addition to its functional capabilities, it also serves as a foundation for future intelligent retail systems. The modular design of the system allows for scalability and the integration of additional features, such as customer tracking, advanced recommendation algorithms, and cloud-based

analytics. This flexibility ensures that the system can evolve alongside technological advancements and changing business requirements.

Overall, the proposed approach demonstrates the potential of combining artificial intelligence and computer vision to transform traditional retail environments into smart, interactive spaces. By enabling real-time customer analysis and personalized content delivery, it contributes to bridging the gap between digital and physical retail experiences, ultimately supporting more effective and engaging customer interactions.

1.1 Background & Literature Review

1.1.1 Age and Gender Detection

Age and gender detection has been widely studied in the field of computer vision, particularly with the rise of deep learning techniques. Early approaches relied on handcrafted features such as facial geometry and texture analysis. However, these methods often struggled with variations in lighting, pose, and image quality.

The introduction of Convolutional Neural Networks (CNNs) significantly improved the accuracy of age and gender prediction. CNN-based models are capable of automatically learning hierarchical features from facial images, making them more robust to variations in real-world conditions [4]. Pretrained models and transfer learning techniques have further enhanced performance by leveraging large-scale datasets [5].

Despite these advancements, challenges remain in achieving consistent accuracy across different age groups, ethnicities, and environmental conditions. Age estimation is inherently difficult due to subtle facial changes over time and individual variations.

1.1.2 Emotion Detection

Emotion recognition systems aim to classify human facial expressions into predefined emotional categories such as happiness, sadness, anger, fear, surprise, and neutrality. Similar to age and gender detection, early methods relied on geometric and appearance-based features extracted from facial landmarks [6].

Modern approaches utilize deep learning models, particularly CNNs, to learn complex patterns associated with facial expressions [7]. These models have demonstrated high accuracy on benchmark datasets. However, real-world emotion detection remains challenging due to factors such as occlusions, lighting variations, cultural differences, and subtle expression changes [8].

Furthermore, emotions are inherently subjective and may not always be accurately represented through facial expressions alone. This introduces uncertainty in classification, especially in dynamic environments such as retail spaces.

1.1.3 Behavioral Engagement Analysis

Behavioral engagement analysis focuses on understanding how users interact with systems or environments. In the context of visual analytics, engagement is often inferred using observable cues such as gaze direction, head orientation, duration of attention, and proximity to the system [3].

Various studies have explored the use of computer vision techniques to estimate engagement levels in applications such as e-learning, human-computer interaction, and advertising. These approaches typically combine multiple indicators to generate an engagement score [9].

However, defining and quantifying engagement remains complex due to its subjective nature. Most existing systems rely on heuristic or rule-based approaches, which may not generalize well across different contexts.

1.1.4 Advertisement Recommendation Systems

Recommendation systems play a crucial role in delivering personalized content to users [10]. Traditional recommendation systems are commonly used in e-commerce and online platforms, where user preferences are inferred from historical data such as browsing behavior and purchase history.

In contrast, real-time recommendation systems in physical environments are less explored. Such systems require immediate analysis of user attributes without relying on historical data. This necessitates the integration of real-time data processing and decision-making mechanisms.

Recent research has highlighted the potential of combining demographic and contextual information to improve recommendation accuracy [1]. However, most systems do not incorporate emotional and engagement-based insights, limiting their effectiveness.

1.1.5 Smart Retail Analytics Systems

Smart retail analytics systems aim to enhance customer experience and business performance by leveraging data-driven insights [2]. These systems often include features such as footfall analysis, heat mapping, and customer segmentation.

While some advanced systems utilize computer vision to track customer movement and behavior, few integrate multiple dimensions such as demographics, emotions, and engagement into a unified framework. Additionally, the direct application of these insights to advertisement personalization remains limited.

1.2 Research Gap

Table 1 : Comparison between existing systems

Feature	Existing Systems (Research)	Proposed Application Component
Age & Gender Detection	Widely researched using CNN-based models with high accuracy; often evaluated in controlled datasets [4]	Integrated real-time age group and gender detection module using live camera input
Emotion Recognition	Focused on facial expression classification using deep learning; limited real-world robustness [7] , [5]	Real-time emotion detection integrated with demographic analysis to provide contextual understanding
Behavioral Engagement Analysis	Limited research; often heuristic-based using gaze or head pose; lacks standardization [3]	Engagement analysis using observable cues such as attention duration, interaction patterns, and presence tracking
System Integration	Most systems focus on a single task (e.g., only emotion or age detection) [7]	Fully integrated pipeline combining age, gender, emotion, and engagement in one system
Real-Time Processing	Many models are tested offline or on static datasets; real-time deployment is limited [2] , [5]	Designed for real-time inference using live video streams in a retail environment
Context-Aware Decision Making	Outputs are often limited to predictions without contextual interpretation [10]	Combines multiple attributes to create meaningful user profiles for decision-making
Advertisement Recommendation	Rarely addressed in research; recommendation systems mostly exist in online platforms [10]	Rule-based advertisement recommendation engine using demographic, emotional, and engagement data

Actionable Outcomes	Systems provide analytical outputs but lack direct application in decision processes [10]	Generates immediate, personalized advertisement outputs based on detected attributes
Retail Environment Application	Limited focus on physical retail implementation; mostly theoretical or lab-based [1]	Specifically designed for deployment in retail kiosks and shopping environments
User Interaction & Experience	Minimal focus on interactive user experience in existing studies [3]	Interactive system (e.g., photobooth/game interface) to increase engagement and data accuracy
Scalability & Deployment	Often not optimized for deployment due to computational complexity [4] , [5]	Built with scalable backend (API-based architecture) suitable for real-world deployment
Commercialization Potential	Research-oriented with limited business application focus [1]	Includes commercialization aspect targeting retail analytics and smart advertising solutions

Another key shortcoming is the absence of a direct linkage between customer analysis and actionable outcomes. Many systems can detect attributes such as age, gender, or emotion; however, these insights are rarely utilized to drive intelligent decision-making processes. In particular, the integration of analytical outputs into a functional recommendation mechanism—such as personalized advertisement delivery—remains underexplored. This disconnect reduces the practical value of these systems, as they stop at analysis rather than enabling meaningful interaction or business impact.

Additionally, limited attention has been given to behavioral engagement as a critical factor in understanding customer interaction. While demographic and emotional attributes provide useful information, engagement indicators such as attention duration, gaze direction, and interaction patterns offer deeper insights into user intent and interest. The lack of incorporation of such behavioral metrics further restricts the effectiveness of existing systems in real-world applications.

Therefore, there is a clear need for a unified, real-time system that not only analyzes multiple customer attributes but also integrates these insights into a coherent decision-making framework. Such a system should bridge the gap between analysis and action by transforming detected attributes into context-aware responses, such as personalized advertisement recommendations. Addressing this gap would contribute significantly to the development of intelligent retail environments that enhance customer experience and improve marketing effectiveness.

1.3 Problem Statement

Physical retail environments lack intelligent systems capable of analyzing customer demographics, emotional states, and behavioral engagement in real time. As a result, advertisements displayed in such environments are typically generic and static, failing to effectively capture the attention of diverse customer groups. This limitation significantly reduces the potential impact of marketing efforts and prevents retailers from delivering personalized experiences comparable to those offered by digital platforms [2].

In contrast to online systems, which utilize extensive user data to tailor content, physical retail spaces often operate with minimal or no real-time customer insights. Retailers are therefore unable to dynamically adapt advertisements based on the characteristics and behavior of customers present at a given moment. This creates a disconnect between customer expectations—shaped by personalized digital experiences—and the reality of in-store interactions [11].

Furthermore, the absence of integrated systems that combine multiple aspects of customer analysis—such as demographic profiling, emotion recognition, and engagement measurement—limits the ability to form a comprehensive understanding of customer behavior. Existing approaches, where available, tend to focus on individual components in isolation and do not provide a unified framework for decision-making [7]. This fragmentation restricts the effectiveness of analytics and prevents the generation of actionable insights.

Another significant challenge lies in the lack of real-time processing and response mechanisms. Even when data is collected, it is often analyzed offline, resulting in delayed insights that cannot influence immediate customer interactions. In retail environments, where customer attention spans are short and decisions are made quickly, the inability to respond in real time further reduces the effectiveness of advertising strategies [12].

Additionally, practical constraints such as varying lighting conditions, multiple users within the frame, occlusions, and diverse facial characteristics introduce complexities in implementing reliable vision-based systems. These challenges make it difficult to deploy existing research solutions directly in real-world environments without significant adaptation [13].

Moreover, there is a lack of direct linkage between customer analysis and actionable outcomes. While certain systems can detect attributes such as age, gender, or emotion, they rarely utilize this information to drive intelligent decisions, such as selecting and displaying personalized advertisements [14]. This disconnect limits the practical value of such systems and reduces their potential contribution to business performance [10].

Therefore, the core research problem addressed in this study is the absence of a unified, real-time intelligent system that can accurately analyze customer demographics, emotions, and engagement, and translate these insights into actionable decisions—specifically, personalized advertisement recommendations—within a physical retail environment.

1.4 Research Objectives

1.4.1 General Objective

To design, develop, and evaluate an intelligent retail analytics system that leverages computer vision and machine learning techniques to analyze customer demographics, emotional states, and behavioral engagement in real time, and to utilize these insights for delivering personalized advertisement recommendations in physical retail environments.

1.4.2 Specific Objectives

- To design and implement a robust age and gender detection module capable of accurately identifying customer demographic attributes from facial images under real-world conditions.
- To develop an emotion recognition system that classifies facial expressions into predefined emotional categories using deep learning techniques.
- To design and implement a behavioral engagement analysis mechanism that estimates user attention and interaction levels using visual cues such as gaze direction, duration of presence, and interaction patterns.
- To develop an intelligent advertisement recommendation system that integrates demographic, emotional, and engagement data to generate context-aware and personalized advertisement outputs.
- To integrate all individual components into a unified real-time system capable of processing live video input and generating immediate recommendations.
- To evaluate the performance, accuracy, and responsiveness of the proposed system using appropriate evaluation metrics and real-time testing scenarios.
- To analyze the practical applicability and limitations of the system within real-world retail environments.

METHODOLOGY

The proposed system is designed as an integrated real-time retail analytics platform that leverages computer vision and machine learning techniques to analyze customer attributes and behavior for personalized advertisement delivery. The system operates by capturing live video input through a camera and processing each frame to extract meaningful insights such as age group, gender, emotional state, and engagement level.

The architecture of the system is structured as a multi-stage processing pipeline. Initially, input frames are captured and preprocessed to detect and isolate facial regions. These detected face regions are then passed through multiple analytical modules, each responsible for extracting specific attributes. The age and gender detection module estimates demographic characteristics, while the emotion detection module identifies the emotional state of the user. In parallel, the behavioral engagement module evaluates user interaction using visual cues such as duration of attention, head orientation, and persistence within the frame.

The outputs from these modules are aggregated to form a comprehensive user profile. This profile is then provided as input to the advertisement recommendation module, which applies predefined decision logic to select contextually relevant advertisements. The selected advertisement is retrieved from a database and displayed through the user interface in real time.

A key feature of the proposed system is its ability to operate under real-time constraints. The system is designed to process continuous video streams efficiently, ensuring minimal latency between input capture and output generation. This enables immediate adaptation of displayed content based on user behavior, which is critical in retail environments where customer attention spans are limited.

Furthermore, the system is implemented using modular architecture, allowing individual components to function independently while contributing to the overall pipeline. This modularity enhances system scalability and flexibility, enabling future integration of additional features such as advanced recommendation algorithms or cloud-based analytics.

Overall, system provides an end-to-end solution that bridges the gap between customer analysis and actionable outcomes by transforming visual data into personalized advertisement recommendations within a physical retail context.

2.1 Requirement Gathering and Analysis

Requirement gathering and analysis is a critical phase in the development of the system, as it defines the functional and operational expectations of the system. This phase involves identifying user needs, system capabilities, and constraints to ensure that the final solution effectively addresses the research problem.

The requirements for the system were derived based on the objectives of developing a real-time retail analytics platform capable of analyzing customer attributes and delivering personalized

advertisements. Inputs for requirement identification were obtained through analysis of retail use cases, system design considerations, and technological feasibility.

The identified requirements are categorized into functional requirements, which define what the system should do, and non-functional requirements, which specify how the system should perform.

2.1.1 Functional Requirements

Functional requirements describe the essential capabilities that the system must perform to achieve its intended objectives. These requirements define the core operations of the system, ensuring that it effectively captures, processes, and utilizes visual data to deliver personalized advertisement recommendations in real time.

Live Video Capture:

At the foundation of the system lies the ability to capture live video input from a camera installed within the retail environment. This continuous video stream serves as the primary data source for all subsequent processing stages. The system must ensure stable and uninterrupted video acquisition to maintain consistent performance.

Face Detection:

Following data acquisition, the system must accurately detect human faces within each video frame. This is a critical step, as all subsequent analyses rely on correctly identifying and isolating facial regions. Face detection ensures that irrelevant background information is filtered out, allowing the system to focus only on meaningful visual data.

Audience Detection and Tracking:

In addition to detecting faces, the system must also be capable of identifying and tracking multiple individuals within the camera's field of view. This functionality is essential for real-world deployment, where multiple customers may interact with the system simultaneously. Effective tracking ensures that the system can maintain continuity in analysis across consecutive frames.

Age and Gender Detection:

Once faces are detected, the system must analyze these regions to extract demographic attributes such as age group and gender. These attributes form a fundamental component of user profiling and play a significant role in tailoring advertisement content to specific audience segments.

Emotion Recognition:

The system must also include the capability to recognize facial expressions and classify them into predefined emotional categories. This allows the system to gain contextual insight into the user's mood, which can significantly enhance the relevance of advertisement recommendations. For example, advertisements can be adapted based on whether a user appears engaged, neutral, or disinterested.

Engagement Analysis:

Beyond demographic and emotional analysis, the system must evaluate user engagement using observable behavioral indicators. These include factors such as duration of presence in front of the display, head orientation, and interaction patterns. By analyzing these cues, the system can estimate the level of user interest and attention, which is crucial for determining the effectiveness of displayed content.

Advertisement Recommendation:

Based on the combined analysis of demographic, emotional, and engagement data, the system must generate personalized advertisement recommendations. This represents the core functionality of system, where analytical outputs are transformed into actionable decisions. The recommendation mechanism must ensure that advertisements are contextually relevant to the detected user profile.

Real-Time Advertisement Display:

To complete the interaction loop, the system must dynamically display selected advertisements in real time. This capability ensures that content is updated immediately in response to changes in user behavior, thereby enhancing user engagement and interaction effectiveness.

Anomaly Detection:

In addition to advertisement delivery, the system may incorporate advanced functionality for detecting abnormal or suspicious behavior. This includes identifying unusual patterns based on predefined thresholds, enabling the system to provide basic security monitoring alongside its primary functionality.

Data Storage:

Furthermore, the system must support data storage capabilities to retain relevant information such as audience metrics, interaction patterns, and advertisement performance. This stored data can be used for further analysis, performance evaluation, and system improvement.

Admin Dashboard:

Finally, the system must provide an administrative interface that allows system operators to monitor performance, analyze collected data, and manage advertisement content. This interface enhances usability and enables effective control over system operations.

2.1.2 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance expectations, and operational constraints of the system. These requirements ensure that the system not only performs its intended functions but also operates efficiently, reliably, and securely in real-world environments.

Real-Time Performance:

A key requirement of the system is its ability to operate in real time. The system must process incoming video frames and generate outputs with minimal delay to ensure immediate responsiveness. Low latency is critical in retail environments, where user attention spans are short and interactions occur within brief time intervals.

Accuracy:

Accuracy is another essential requirement, particularly for the machine learning components of the system. The system must maintain a high level of accuracy in detecting age, gender, and emotional states to ensure that advertisement recommendations are reliable and meaningful. Inaccurate predictions may lead to irrelevant content, reducing system effectiveness.

Scalability:

The system must also be scalable, allowing it to handle multiple users and operate across different deployment environments. As retail spaces vary in size and complexity, the system should be adaptable to both small kiosks and large-scale installations without significant performance degradation.

Reliability:

Reliability is crucial for continuous operation. The system must function consistently under varying real-world conditions, including changes in lighting, multiple users within the frame, and partial occlusions of faces. Robustness against such variations ensures stable performance in practical scenarios.

Usability:

From a usability perspective, the system should provide an intuitive and user-friendly interface. Both end users and administrators should be able to interact with the system easily, without requiring extensive technical knowledge. A well-designed interface enhances user experience and system adoption.

Maintainability:

Maintainability is another important consideration. The system should be designed using modular architecture, allowing individual components to be updated, modified, or replaced without affecting the entire system. This facilitates easier debugging, upgrades, and integration of new features.

Security:

Security requirements ensure that system data is protected against unauthorized access. Proper mechanisms must be in place to safeguard stored data and prevent misuse. This is particularly important when handling data generated from real-world interactions.

Privacy:

Privacy considerations are also critical. The system should avoid storing personally identifiable information and should operate in a manner that respects user anonymity. Ethical guidelines must be followed to ensure responsible use of technology.

Compatibility:

Compatibility is required to ensure that the system can operate with commonly available hardware such as webcams, display screens, and standard computing devices. This enables easier deployment across different retail environments.

Performance Efficiency:

Finally, the system must be efficient in terms of resource utilization. It should optimize the use of computational resources such as CPU and GPU to ensure smooth operation without requiring excessive hardware capabilities. Efficient performance is particularly important for real-time systems operating continuously.

2.2 System Architecture

The system architecture of system is designed as a multi-layered, modular pipeline that enables real-time processing of visual data for intelligent advertisement delivery and anomaly detection. The architecture is structured into four primary layers: Input Layer, AI Processing Layer, Decision Layer, and Data & Management Layer, as illustrated in Figure 1.

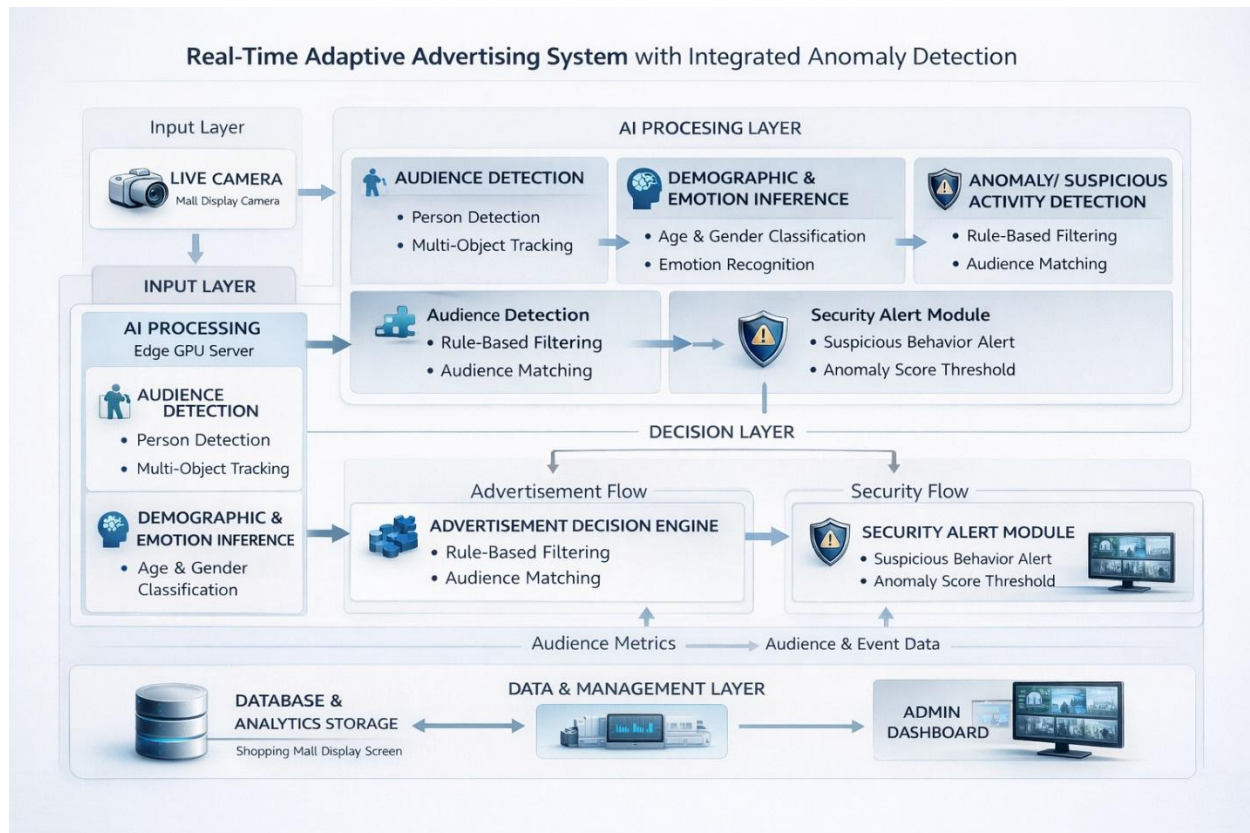


Figure 1: Overall system Diagram

2.2.1 Input Layer

The system begins with the input layer, where live video data is captured using a mall display camera. This camera continuously streams real-time video frames representing the environment in front of the digital display.

The primary purpose of this layer is to provide a continuous stream of visual data to the system. Since the system operates in real time, the quality and consistency of input data are critical for accurate analysis. The captured frames are forwarded to the AI processing layer for further analysis.

2.2.2 AI Processing Layer

The AI processing layer is the core component of the system, responsible for extracting meaningful insights from raw visual data. This layer operates on an edge-based processing setup (Edge GPU Server), ensuring low latency and real-time responsiveness.

This layer consists of several interconnected modules:

a) Audience Detection Module

The audience detection module is responsible for identifying and tracking individuals present in the camera frame. It performs:

- Person Detection – Identifies human presence within the frame
- Multi-Object Tracking – Tracks multiple individuals across frames

This module ensures that only relevant human subjects are processed further, filtering out irrelevant background information.

b) Demographic and Emotion Inference Module

Once individuals are detected, their facial regions are analyzed to extract demographic and emotional attributes. This module performs:

- Age and Gender Classification – Predicts age group and gender using facial features
- Emotion Recognition – Identifies emotional states such as happiness, sadness, or neutrality

These attributes provide essential contextual information about the user, which is later used in decision-making.

c) Audience Filtering and Matching

This sub-module applies rule-based filtering mechanisms to refine detected users. It ensures:

- Removal of low-confidence detections
- Selection of primary target audience
- Matching users to predefined audience segments

This step improves system reliability by focusing only on relevant and high-confidence detections.

d) Anomaly / Suspicious Activity Detection Module

In addition to advertisement personalization, the system includes a security-focused component that detects abnormal or suspicious behavior.

This module performs:

- Rule-Based Filtering for anomalies
- Audience behavior analysis
- Suspicious activity identification based on thresholds

This enables the system to serve dual purposes:

1. Advertisement optimization

2. Security monitoring

e) Security Alert Module

When abnormal behavior is detected, the system triggers alerts using:

- Suspicious Behavior Alerts
- Anomaly Score Threshold Evaluation

This ensures real-time monitoring and enhances safety in retail environments

2.2.3 Decision Layer

The decision layer acts as the intelligence hub of the system, where processed data is transformed into actionable outputs. It consists of two main flows:

a) Advertisement Flow

The advertisement decision engine is responsible for selecting appropriate advertisements based on analyzed user attributes.

It uses:

- Demographic data (age, gender)
- Emotional state
- Engagement level

A rule-based logic is applied to match user profiles with relevant advertisement categories. This ensures that the displayed content is personalized and contextually relevant.

b) Security Flow

Parallel to the advertisement system, the security flow handles anomaly detection outputs. When suspicious activity is identified, alerts are generated and displayed through the monitoring interface.

This dual-flow design enhances the system's functionality by combining marketing intelligence with security monitoring.

2.2.4 Data and Management Layer

The data and management layer is responsible for storing, managing, and visualizing system outputs.

a) Database and Analytics Storage

All relevant data, including:

- Audience metrics
- Event data
- Advertisement performance

are stored in a centralized database. This allows for historical analysis and performance tracking.

b) Data Processing and Management

This component processes stored data to generate insights such as:

- Audience trends
- Engagement statistics
- Advertising effectiveness

c) Admin Dashboard

The admin dashboard provides a user interface for monitoring system performance. It allows administrators to:

- View live analytics
- Monitor security alerts
- Analyze audience behavior

This enhances usability and enables data-driven decision making for retailers.

2.3 Data Collection and Preprocessing

2.3.1 Data Collection and Sources

Data collection forms a fundamental component of the system, as the accuracy and reliability of the system heavily depend on the quality and diversity of the data used for training and evaluation. The system utilizes a combination of pretrained models, publicly available datasets, and real-time video input to ensure both robustness and practical applicability.

For demographic analysis, age and gender detection models are developed using labeled facial image datasets that contain annotations for age groups and gender categories. These datasets typically include a wide range of facial variations in terms of age, ethnicity, lighting conditions, and facial orientations, enabling the model to generalize effectively in real-world scenarios.

Similarly, the emotion detection module is trained using datasets that consist of facial expressions categorized into predefined emotional classes such as happiness, sadness, anger, surprise, fear, and

neutrality. These datasets provide the necessary diversity for the model to learn distinct facial patterns associated with different emotional states.

In addition to offline datasets, the system processes real-time video input captured through a camera installed in the retail environment. This live input serves as the primary data source during system operation, allowing the system to analyze customer behavior dynamically. The inclusion of real-time data enables the evaluation of system performance under practical conditions, including varying lighting, multiple users, and environmental noise.

The combination of pretrained models and real-time input ensures that the system is both accurate and adaptable. Pretrained models provide strong baseline performance, while live data enables continuous interaction and real-world validation of the system.

2.3.2 Data Preprocessing

Data preprocessing is a critical stage in the system, as it transforms raw visual input into a structured format suitable for machine learning analysis. Since the system operates on continuous video streams, preprocessing ensures that relevant information is extracted efficiently while minimizing noise and inconsistencies. The preprocessing pipeline consists of several sequential steps designed to standardize input data and improve model performance.

Frame Acquisition and Sampling

The system captures video input in real time and converts it into individual frames for processing. To maintain real-time performance and avoid excessive computational load, frames are processed at controlled intervals rather than analyzing every single frame. This ensures a balance between processing speed and analytical accuracy.

Face Detection and Localization

Once frames are extracted, the system identifies human faces using the Multi-task Cascaded Convolutional Neural Network (MTCNN). This step is essential as it isolates facial regions from the background, ensuring that subsequent analysis focuses only on relevant visual information.

The detection process includes identifying bounding boxes around faces and extracting key facial landmarks. Low-confidence detections are filtered out to maintain reliability.

Face Cropping and Alignment

After detection, facial regions are cropped from the original frame. This removes unnecessary background information and allows the system to focus on critical facial features.

In some cases, alignment techniques are applied to standardize face orientation. This helps reduce variations caused by head tilts or angles, thereby improving the accuracy of downstream models.

Image Resizing

The cropped facial images are resized to match the input dimensions required by the machine learning models. Standardizing image size ensures consistency across inputs and reduces computational complexity during processing.

Normalization

Pixel values of the images are normalized to a standard range, typically between 0 and 1 or -1 and 1. This improves numerical stability and ensures that the models can process input data efficiently.

Noise Reduction and Quality Filtering

Real-world video data often contains noise due to lighting conditions, motion blur, or camera limitations. To address this, the system filters out low-quality frames and applies basic enhancements where necessary.

Faces that do not meet predefined quality or confidence thresholds are discarded to prevent inaccurate predictions.

Handling Real-World Variations

The preprocessing pipeline is designed to handle challenges commonly encountered in real-world environments, such as:

- Variations in lighting conditions
- Presence of multiple users in the frame
- Partial occlusions of faces
- Differences in distance from the camera

By addressing these challenges, the system maintains robustness and reliability during operation.

Data Flow to Analytical Modules

After preprocessing, the refined facial data is forwarded to the analytical modules of the system, including:

- Age and gender detection
- Emotion recognition
- Engagement analysis

Standardized inputs ensure that all modules receive consistent data, enabling efficient and accurate processing. The effectiveness of the system is highly dependent on the quality of data collection and preprocessing. Accurate datasets enable the development of reliable models, while efficient preprocessing ensures that real-time data is handled effectively. By combining diverse training datasets with real-time input and applying robust preprocessing techniques, the system achieves a balance between accuracy, efficiency, and practical usability. This integration is essential for delivering reliable performance in dynamic retail environments.

2.4 Model Design

The system integrates multiple machine learning models to extract meaningful insights from facial data in real time. The primary analytical components include the age and gender detection model and the emotion recognition model, both of which are based on deep learning techniques.

These models are designed to process preprocessed facial images and generate predictions that contribute to the overall user profile. The outputs from these models are subsequently used in the advertisement recommendation process.

The design of these models focuses on achieving a balance between accuracy, computational efficiency, and real-time performance, which is essential for deployment in retail environments.

2.4.1 Age and Gender Detection Model

a) Objective of the Model

The primary objective of the age and gender detection model is to classify the demographic attributes of users based on facial features. This includes predicting:

- Age group (e.g., 0–12, 13–19, 20–39, 40–59, 60+)
- Gender (Male / Female)

These attributes are critical for identifying target audience segments and enabling personalized advertisement delivery.

b) Model Architecture

The age and gender detection module is based on a Convolutional Neural Network (CNN) architecture. CNNs are widely used in image-based tasks due to their ability to automatically extract hierarchical features from input images [15].

The architecture typically consists of:

- Convolutional layers for feature extraction
- Activation functions (ReLU) to introduce non-linearity
- Pooling layers to reduce spatial dimensions
- Fully connected layers for classification
- Output layer with softmax activation

The model is designed to process facial images resized to a fixed input size (e.g., 128×128 pixels), ensuring consistency in input data.

c) Input and Output

- Input: Preprocessed facial image
- Output:

- Age group classification
- Gender classification

The model may use either a multi-output architecture (predicting both age and gender simultaneously) or separate output layers for each task [16].

d) Training Process

The model is trained using labeled facial image datasets containing annotations for age and gender [17]. The training process includes:

- Data preprocessing and augmentation
- Forward propagation through CNN
- Loss calculation using appropriate loss functions (e.g., categorical cross-entropy)
- Backpropagation and weight updates

Optimization techniques such as Adam optimizer are used to improve convergence speed and accuracy.

e) Model Optimization

To ensure real-time performance, the model is optimized by:

- Reducing model complexity
- Using lightweight architecture
- Applying efficient inference techniques

This ensures that predictions are generated quickly without compromising system responsiveness.

f) Limitations

Despite its effectiveness, the model may face challenges such as:

- Difficulty in predicting exact age due to facial variations
- Bias in datasets affecting generalization
- Sensitivity to lighting and image quality

2.4.2 Emotion Detection Model

a) Objective of the Model

The emotion detection model is designed to classify facial expressions into predefined emotional categories. The purpose of this model is to provide contextual understanding of user mood, which enhances the personalization of advertisements.

b) Model Architecture

Similar to the demographic model, the emotion detection system is based on a Convolutional Neural Network (CNN) architecture [18].

The model consists of:

- Multiple convolutional layers for feature extraction
- Activation layers (ReLU)
- Pooling layers to reduce dimensionality
- Dropout layers to prevent overfitting
- Fully connected layers for classification

The final layer uses a softmax activation function to produce probability scores for each emotion class.

c) Input and Output

- Input: Preprocessed facial image
- Output: Probability distribution across emotion classes

The emotion with the highest probability is selected as the final prediction.

d) Training Process

The model is trained using datasets containing labeled facial expressions. The training process includes [19]:

- Data normalization
- Augmentation (rotation, flipping, brightness adjustments)
- Loss function optimization using categorical cross-entropy
- Model validation to prevent overfitting

e) Real-Time Inference

During real-time operation, the model processes incoming frames and generates predictions continuously. To improve stability, predictions may be smoothed across multiple frames to avoid rapid fluctuations.

f) Challenges

Emotion detection is inherently complex due to:

- Subtle variations in facial expressions
- Cultural differences in emotional representation
- Occlusions such as masks or accessories

These challenges can impact model accuracy in real-world scenarios.

2.4.3 Behavioral Engagement Analysis

Behavioral engagement analysis is a key component of the system, as it provides insight into how users interact with the retail display. While demographic and emotional attributes describe who the user is and how they feel, engagement analysis determines the level of user attention and interest, which is essential for delivering effective advertisement recommendations [20].

In the proposed system, engagement is estimated using a combination of computer vision-based feature extraction and model-assisted inference techniques, enabling a more robust and data-driven approach compared to simple rule-based methods.

a) Objective of the Model

The primary objective of engagement analysis is to quantify user interaction with the system based on observable behavioral cues. These cues help determine whether a user is:

- Actively engaged
- Passively observing
- Not interested

This classification enables the system to adapt advertisement delivery accordingly.

b) Model Architecture

The engagement analysis module utilizes multiple features extracted from the video stream. These include:

- Face Presence Duration: The amount of time a user remains within the camera frame
- Head Orientation: Indicates whether the user is looking towards the display
- Face Detection Consistency: Frequency of detection across consecutive frames
- Distance from Camera (optional): Approximation of user proximity
- Motion Stability: Indicates whether the user is stationary or moving away

These features are derived from outputs of the preprocessing and detection modules.

c) Training Process

To improve robustness, engagement is estimated using a lightweight predictive model combined with temporal aggregation, rather than relying solely on fixed thresholds.

The system utilizes:

- A temporal scoring mechanism, where engagement is calculated across multiple frames
- A feature-based scoring model, which assigns weights to different behavioral indicators

The resulting score is normalized and categorized into:

- High Engagement
- Medium Engagement
- Low Engagement

Temporal Smoothing

Since real-time predictions can fluctuate between frames, the system applies temporal smoothing techniques to stabilize engagement estimation.

This includes:

- Sliding window averaging
- Majority voting across recent frames
- Probability aggregation

This ensures consistent and reliable engagement predictions.

Incorporating engagement analysis significantly enhances system effectiveness by:

- Differentiating between active and passive users
- Improving advertisement targeting accuracy
- Increasing interaction efficiency

Without engagement analysis, the system would rely solely on static attributes, reducing its ability to respond dynamically to user behavior.

Integration with Other Models

The engagement module operates in conjunction with:

- Age and gender detection model
- Emotion recognition model

The combined output forms a multi-dimensional user profile, which improves decision-making in the recommendation stage.

2.4.4 Advertisement Recommendation System

The advertisement recommendation system is the core decision-making component of system. It transforms analytical outputs generated by machine learning models into actionable outcomes by selecting and displaying personalized advertisements.

a) Objective of the Model

The primary objective of the recommendation system is to deliver context-aware advertisements tailored to individual users based on:

- Demographic attributes (age, gender)
- Emotional state
- Engagement level

This ensures that advertisements are relevant, timely, and effective.

b) Input Features

The recommended engine receives input from multiple analytical modules:

- Age group (from demographic model)
- Gender (from demographic model)
- Emotion (from emotion detection model)
- Engagement level (from engagement module)

These inputs collectively define the user profile.

c) Recommendation Logic

The system utilizes a hybrid rule-based decision model supported by machine learning outputs [21].

The logic operates as follows:

1. Identify user demographic segment
2. Evaluate emotional context
3. Adjust based on engagement level
4. Select matching advertisement category

For example:

- Young + High engagement + Happy → Entertainment ads
- Adult + Neutral + Medium engagement → Lifestyle ads
- Low engagement → Attention-grabbing ads

d) Model-Assisted Recommendation

Although the system primarily uses rule-based mapping, it is enhanced by model outputs, making it a model-assisted decision system.

Future improvements may include:

- Reinforcement Learning Based recommendations
- User feedback-driven optimization

- Adaptive learning from advertisement performance

e) Advertisement Database

The system maintains a database of advertisements categorized based on:

- Target age group
- Gender preference
- Emotional relevance
- Advertisement type (image/video)

Each advertisement is tagged with metadata, enabling efficient retrieval during runtime.

2.4.5 Real-Time Advertisement Selection

The recommendation system operates in real time [22], continuously updating displayed advertisements based on:

- Changes in user profile
- Engagement fluctuations
- Multiple user scenarios

This ensures dynamic and responsive content delivery.

Handling Multiple Users

In scenarios with multiple users, the system:

- Identifies the dominant user (based on engagement)
- Applies weighted decision-making
- Selects advertisements targeting the majority profile

The recommendation system plays a critical role in:

- Converting analysis into actionable output
- Enhancing advertisement effectiveness
- Improving customer engagement

It represents the final stage of the pipeline where data-driven insights are translated into real-world impact.

2.5 Commercialization Aspect of the Product

The system demonstrates strong potential for commercialization as an intelligent retail analytics and advertisement recommendation platform. By combining real-time computer vision with

machine learning-based decision-making, the system addresses a critical gap in physical retail environments, where personalization and data-driven marketing are currently limited.

2.5.1 Market Opportunity

The retail industry is increasingly shifting towards data-driven strategies to enhance customer engagement and improve marketing effectiveness. While online platforms provide highly personalized experiences through user data analytics, physical retail environments still rely on static advertisements and manual observation.

System addresses this gap by enabling real-time customer analysis and personalized advertisement delivery. The system is particularly suitable for deployment in:

- Shopping malls
- Supermarkets and retail stores
- Digital advertising kiosks
- Airports and transportation hubs
- Exhibition centers and public spaces

These environments present high foot traffic and diverse customer groups, making them ideal for intelligent advertisement systems.

2.5.2 Value Proposition

The primary value proposition of system lies in its ability to transform traditional advertising into a dynamic and personalized experience. Key benefits include:

- Enhanced Customer Engagement: Delivering advertisements tailored to user demographics, emotions, and behavior
- Improved Marketing Effectiveness: Increasing the relevance of advertisements and potential conversion rates
- Real-Time Decision Making: Adapting content instantly based on user interaction
- Actionable Insights: Providing retailers with data on customer behavior and engagement patterns

This combination of features enables businesses to optimize their advertising strategies and improve return on investment (ROI).

2.5.3 Business Model

System can be commercialized using multiple business models depending on deployment scale and target market.

a) Software-as-a-Service (SaaS)

The system can be offered as a subscription-based platform where retailers pay a monthly or annual fee for access to:

- Real-time analytics
- Advertisement recommendation system
- Performance dashboards

b) Hardware + Software Bundle

System can be deployed as a complete solution, including:

- Camera setup
- Processing unit (edge device)
- Display system
- Software platform

This model is suitable for malls and retail chains looking for a ready-to-use solution.

c) Data Analytics Service

Retailers can subscribe to analytics services that provide:

- Customer behavior reports
- Engagement metrics
- Advertisement performance analysis

This adds value beyond advertisement display by supporting strategic decision-making.

2.5.4 Scalability and Deployment

The system is designed with scalability in mind, allowing deployment across multiple locations. Its modular architecture enables:

- Integration with existing retail infrastructure
- Expansion to multiple displays and cameras
- Centralized monitoring through dashboards

System can be deployed in:

- Edge environments for real-time processing
- Cloud environments for large-scale data storage and analytics

This flexibility supports both small-scale and enterprise-level implementations.

2.5.5 Competitive Advantage

The system offers several significant advantages over traditional and existing smart retail systems by combining multiple intelligent capabilities into a unified platform. Unlike conventional systems that focus only on basic customer tracking or static advertising, integrates demographic analysis

(such as age and gender), emotion detection, and customer engagement monitoring within a single system. This holistic approach enables retailers to gain deeper insights into customer behavior and preferences, allowing for more personalized and effective decision-making.

One of the key strengths of this system is its real-time processing capability. The system can instantly analyze customer data as they interact with the environment, enabling dynamic advertisement adaptation. For example, advertisements can be adjusted on-the-fly based on the detected audience profile or emotional response, significantly increasing marketing effectiveness and customer satisfaction.

Additionally, this provides dual functionality by combining smart advertising with basic security monitoring. While delivering targeted advertisements, the system can also detect unusual behavior patterns, contributing to safety and surveillance within the retail environment.

The modular and scalable design of system further enhances its practicality. Each component—such as emotion detection, engagement tracking, or recommendation systems—can function independently or be expanded as needed. This flexibility allows businesses to adopt the system according to their size, budget, and specific requirements, making it suitable for both small retail stores and large shopping malls.

Overall, these features clearly differentiate from traditional advertising platforms. By combining intelligent analytics, real-time adaptability, and multifunctional capabilities, the system provides a strong competitive edge in the evolving smart retail market, enabling businesses to optimize both customer engagement and operational efficiency.

2.5.6 Challenges and Considerations

Despite its potential, several challenges must be considered for commercialization:

- Privacy and Ethical Concerns: Ensuring compliance with data protection regulations
- Initial Setup Cost: Hardware and infrastructure requirements
- Environmental Variability: Performance under different lighting and crowded conditions

Addressing these challenges is essential for successful adoption.

2.5.7 Future Commercial Enhancements

To further enhance commercialization potential, the system can be extended with:

- AI-driven adaptive recommendation algorithms
- Integration with sales and transaction data
- Cloud-based analytics dashboards
- Multi-language support for global deployment
- Advanced user segmentation and targeting

These improvements can increase system value and expand its market reach.

2.6 Testing & Implementation

2.6.1 Implementation

Tools and Technologies

The development of the system involves the integration of multiple software frameworks, programming languages, and hardware components to enable real-time computer vision processing and intelligent decision-making. The selected tools and technologies are chosen based on their efficiency, scalability, and suitability for handling real-time data processing tasks.

Programming Language

The primary programming language used in the development of the system is Python. Python is widely adopted in the field of artificial intelligence and computer vision due to its simplicity, extensive library support, and strong community backing. It enables rapid development and seamless integration of machine learning models and APIs.

Computer Vision Library

OpenCV (Open-Source Computer Vision Library) is used for handling image and video processing tasks. OpenCV provides functionalities such as:

- Video frame capture
- Image manipulation
- Face detection support
- Real-time processing

Its efficiency and real-time capabilities make it suitable for handling continuous video streams in the system.

Face Detection Framework

The system utilizes Multi-task Cascaded Convolutional Neural Network (MTCNN) for face detection. MTCNN is a deep learning-based algorithm capable of detecting faces and facial landmarks with high accuracy.

It is used to:

- Detect face bounding boxes
- Extract facial regions
- Identify key facial landmarks

This ensures accurate input for subsequent machine learning models.

Machine Learning Frameworks

The machine learning components of the system are implemented using:

- TensorFlow / Keras
- ONNX Runtime

TensorFlow, along with its high-level API Keras, is utilized for the design, training, and evaluation of deep learning models. These frameworks provide extensive support for building Convolutional Neural Networks (CNNs), which are particularly well-suited for image-based tasks such as age, gender, and emotion detection. The availability of prebuilt layers, optimization algorithms, and training utilities enables efficient model development and experimentation.

ONNX Runtime is employed during the deployment phase to perform optimized inference of trained models. By converting trained models into the ONNX (Open Neural Network Exchange) format, the system achieves faster execution and improved compatibility across different environments. This is particularly important for real-time applications, where low latency and high throughput are critical.

Deep Learning Models

The system incorporates multiple deep learning models to perform facial analysis tasks. These include models for age and gender detection as well as emotion recognition. Each model is based on Convolutional Neural Network (CNN) architectures, which can extract hierarchical features from facial images.

These models are designed to operate efficiently in real-time conditions, ensuring that predictions are generated with minimal delay. Optimization techniques such as lightweight architectures and efficient inference frameworks are applied to maintain a balance between accuracy and computational performance.

The outputs of these models form the core analytical input for the system, enabling the generation of personalized advertisement recommendations.

Backend Framework

The backend of the system is developed using FastAPI, a modern web framework for building high-performance APIs, a modern and high-performance web framework for building APIs. FastAPI is chosen for its ability to handle asynchronous operations efficiently, making it suitable for real-time applications that require rapid data processing and response generation.

The backend is responsible for managing the overall data flow within the system. It handles incoming requests from the frontend, processes video frames, performs model inference, and returns the results to the user interface. Additionally, FastAPI enables seamless integration

between machine learning models and the frontend application, ensuring smooth communication across system components.

Frontend Framework

The user interface of the system is developed using React, a widely used JavaScript library for building dynamic and interactive web applications. React provides a component-based architecture that allows efficient management of user interface elements and real-time updates.

In the system, React is used to display advertisements dynamically based on system outputs. It enables smooth rendering of visual content and provides interactive user experience. The framework also supports efficient state management, ensuring that the interface responds quickly to changes in user data and system predictions.

Database System

The system utilizes MongoDB, a NoSQL database, for storing and managing data generated during system operation. MongoDB is particularly suitable for this application due to its flexibility in handling unstructured and semi-structured data.

The database stores various types of information, including advertisement content, user interaction logs, and system outputs. This data can be used for performance analysis, system monitoring, and future enhancements such as data-driven recommendation improvements.

MongoDB's scalability and high-performance capabilities make it an appropriate choice for applications that may expand to handle large volumes of real-time data.

Hardware Components

The system operates using standard hardware components, including:

- Webcam or camera for video input
- Processing unit (CPU/GPU) for model inference
- Display screen for advertisement output

In some cases, GPU acceleration may be used to improve performance in real-time processing.

Deployment Environment

The system is designed to be flexible and adaptable to different deployment environments. It can be deployed on local machines for development and testing purposes, allowing rapid iteration and debugging.

For real-time operation in retail environments, the system can be deployed on edge devices, where processing is performed close to the data source. This reduces latency and ensures faster response times.

Additionally, the system can be integrated with cloud platforms to support large-scale data storage, analytics, and centralized monitoring. Cloud deployment enables scalability and facilitates advanced features such as remote management and performance tracking.

This flexibility in deployment ensures that the system can be adapted to a wide range of retail scenarios, from small kiosks to large commercial environments.

Table 2: Summary of Tools and Technologies

Component	Technology
Programming Language	Python
ML Framework	TensorFlow / Keras
Inference Engine	ONNX Runtime
Backend	FastAPI
Frontend	React
Database	MongoDB
Vision	OpenCV
Face Detection	MTCNN
People Tracking	SORT

2.6.2 Testing and Evaluation

Testing and evaluation are critical processes in validating the effectiveness, reliability, and real-time performance of the system. Since the system integrates multiple machine learning components such as age and gender detection, emotion recognition, and engagement analysis, it is essential to evaluate both individual modules and the complete system under realistic operational conditions. The evaluation strategy is designed to ensure that the system satisfies both functional requirements, such as accurate prediction, and non-functional requirements, such as responsiveness and scalability.

Module-Level Testing

At the module level, each component is tested independently to verify its functionality and predictive accuracy before integration into the full system.

a) Age and Gender Detection Testing

The age and gender detection module is evaluated using labeled facial image datasets, where the model's performance is measured primarily through classification accuracy. In addition to accuracy, consistency across multiple frames is examined to ensure stable predictions over time. The model is tested under a variety of conditions, including different lighting environments, variations in facial orientation, and diverse age groups, to assess its robustness and generalization capability.

b) Emotion Detection Testing

The emotion detection module is evaluated using datasets containing labeled emotional expressions. The focus of this evaluation is not only on classification accuracy but also on the model's ability to distinguish between subtle facial expressions that may appear similar. Real-world challenges such as occlusions caused by glasses or masks and rapid changes in facial expressions are also considered during testing. This ensures that the model remains reliable in dynamic and uncontrolled environments.

c) Engagement Analysis Testing

The engagement analysis module is tested using controlled scenarios that simulate user interaction with the system. This module analyzes behavioral patterns such as the duration of user presence, attention direction, and interaction intensity. A key aspect of this evaluation is the consistency of engagement scores across consecutive frames, as unstable predictions could reduce the reliability of the system. By ensuring temporal stability, the system can provide meaningful insights into user engagement.

System-Level Testing

After validating individual modules, system-level testing is conducted to evaluate the performance of the integrated pipeline. This phase ensures that data flows correctly between components and that outputs generated by one module are accurately utilized by others. For instance, demographic and emotional data must seamlessly feed into the advertisement recommendation module to generate relevant outputs.

The system is tested using real-time video input, often involving multiple users, to replicate real-world retail environments. This helps verify that the system can handle complex scenarios where multiple variables interact simultaneously.

Real-Time Performance Evaluation

Given that system is designed as a real-time intelligent system, performance evaluation places significant emphasis on responsiveness and efficiency. One of the primary metrics considered is latency, which refers to the time delay between capturing input data and generating the

corresponding output. Lower latency is essential to ensure that advertisements are displayed without noticeable delay.

In addition, the frame processing rate is evaluated to determine how many frames the system can process per second, which directly impacts the smoothness of operation. System responsiveness is also assessed by observing how quickly the system adapts to changes in user behavior, such as movement or variation in engagement levels. Optimization techniques are applied to balance computational efficiency with prediction accuracy.

Evaluation Metrics

To quantitatively measure system performance, several evaluation metrics are used:

a) Accuracy

Accuracy is used to evaluate classification performance of the models:

- Age classification accuracy
- Gender classification accuracy
- Emotion classification accuracy

b) Precision and Recall

These metrics may be used for deeper analysis of classification performance:

- Precision: Correct positive predictions
- Recall: Ability to detect all relevant instances

c) Latency

Latency measures the delay between capturing input and generating output. Lower latency indicates better real-time performance. A system with high latency may fail to deliver timely recommendations, reducing its effectiveness in a retail setting.

d) Consistency

Consistency measures how stable predictions are across consecutive frames. This is particularly important for emotion and engagement analysis. This is particularly important for emotion detection and engagement analysis, where sudden fluctuations can lead to unreliable outputs.

Real-World Testing Scenarios

These tests ensure that the system performs reliably in practical deployment scenarios. To ensure practical applicability, the system is tested under various real-world conditions. These include environments with different lighting intensities, scenarios involving multiple users within the same frame, and situations where users enter and exit the scene dynamically. Additionally, the system is

evaluated under conditions where faces are partially occluded, which is a common occurrence in real-world settings. These tests help verify the robustness and adaptability of the system in operational environments.

Challenges Observed During Testing

Despite achieving satisfactory performance, several challenges are observed during testing. The accuracy of detection tends to decrease under poor lighting conditions, highlighting the sensitivity of computer vision models to environmental factors. Similarly, extreme facial angles can affect the reliability of both demographic and emotion predictions. Emotion detection can be inconsistent when expressions are subtle or ambiguous. Another challenge arises when multiple users are present in the frame, as this increases computational complexity and may impact real-time performance. These observations indicate areas where further optimization and model refinement are required.

System Validation

The overall system is validated based on:

- Successful detection of user attributes
- Correct advertisement recommendation
- Stable real-time performance

The results indicate that the system can function effectively as an intelligent retail analytics solution. The overall system is validated by assessing its ability to perform key tasks effectively. This includes accurate detection of user attributes, correct classification of emotional states, and generation of relevant advertisement recommendations. In addition, the system's ability to maintain stable real-time performance is evaluated. The results demonstrate that system successfully integrates multiple intelligent components into a cohesive system capable of enhancing retail analytics and customer engagement.

Testing and evaluation confirm that the system performs effectively in both controlled and real-world conditions. The system demonstrates reliable performance in detecting user attributes and generating personalized advertisements in real time.

The evaluation highlights the strengths of the system while also identifying areas for improvement, providing a foundation for future enhancements.

RESULTS & DISCUSSION

3.1 Results

Results of Age and Gender Detection

The age and gender detection module was evaluated using validation datasets and real-time input data. The model demonstrated strong performance in identifying gender, with higher accuracy compared to age classification.

Age classification showed acceptable performance, particularly when predicting broader age groups. However, variations were observed in cases involving:

- Extreme lighting conditions
- Partial occlusions
- Subtle facial differences

In contrast, age classification demonstrated moderate but acceptable performance. The model was more reliable when predicting broader age categories rather than exact age values, which aligns with the inherent complexity of age estimation. Variations in prediction accuracy were observed under challenging conditions such as extreme lighting, partial occlusions, and subtle facial differences between individuals. These factors introduce noise and reduce feature clarity, making accurate age prediction more difficult.

Despite these limitations, the model consistently categorizes users into appropriate age groups, which is sufficient for the intended purpose of advertisement targeting. From an application perspective, approximate age grouping is more valuable than precise estimation, as it supports meaningful segmentation of users for personalized content delivery.

Discussion

The evaluation highlights that gender prediction is a relatively straightforward task due to clearly distinguishable facial characteristics, whereas age prediction remains inherently complex and sensitive to environmental and biological variations. The decision to use grouped age categories significantly improved classification stability and reduced prediction errors. This design choice reflects a practical balance between model complexity and real-world usability, ensuring that the system delivers reliable outputs for downstream applications.

Results of Emotion Detection

The emotion detection module was evaluated using labeled datasets along with real-time video streams to measure its effectiveness in dynamic environments. The results show that the model performs well in identifying commonly expressed emotions such as happiness and neutrality, which typically exhibit clear and distinguishable facial patterns.

However, the model showed comparatively lower accuracy when classifying emotions such as fear and surprise. These emotions often involve subtle and overlapping facial features, making them more difficult to distinguish. During real-time testing, fluctuations in predictions were observed across consecutive frames, particularly when users displayed rapid or ambiguous changes in expression.

Discussion

Emotion recognition is inherently challenging due to the subjective and dynamic nature of human facial expressions. Unlike demographic attributes, emotions can change rapidly and may not always be clearly expressed. To address this limitation, temporal smoothing techniques were applied, allowing the system to consider predictions across multiple frames rather than relying on single-frame outputs. This approach improved the stability and reliability of predictions.

Although minor inaccuracies remain, the emotion detection module provides valuable contextual insights that enhance the personalization of advertisements. The inclusion of emotional data allows the system to move beyond static targeting and adapt content based on user mood and engagement level.

Results of Behavioral Engagement Analysis

The engagement analysis module was evaluated through real-time observation of user interaction with the system. The system successfully categorized engagement levels based on:

- Duration of presence
- Head orientation
- Consistency of face detection

Users who remained in front of the display for extended periods and maintained visual focus were classified as highly engaged. In contrast, users who briefly passed by or did not exhibit sustained attention were categorized as having low engagement. The system demonstrated the ability to distinguish between active and passive users effectively.

Discussion

The engagement analysis model proved to be effective in capturing user interest through behavioral patterns rather than relying solely on static features. The use of temporal scoring mechanisms enhanced reliability by reducing fluctuations caused by frame-level inconsistencies. While engagement estimation cannot be perfectly precise due to variability in human behavior, it provides a meaningful approximation that significantly improves decision-making within the recommendation system. This capability is particularly important in identifying users who are more likely to respond to targeted advertisements.

Results of Advertisement Recommendation System

The advertisement recommendation system was evaluated by analyzing the relevance of selected advertisements based on user profiles.

The system successfully generated context-aware recommendations by combining:

- Age group
- Gender
- Emotion
- Engagement level

The results demonstrate that the system can effectively translate analytical outputs into actionable decisions. For instance, younger users exhibiting high engagement were typically shown entertainment-related advertisements, while adults displaying neutral expressions were presented with lifestyle or product-oriented content. These recommendations align with expected marketing strategies, indicating that the system produces logically consistent outputs.

Discussion

The rule-based recommendation approach provides a clear and explainable mechanism for decision-making, which is advantageous during the initial stages of system deployment. It ensures predictable behavior and allows easy interpretation of results. However, the system currently lacks adaptability, as it does not learn from user interactions over time. Incorporating adaptive or machine learning-based recommendation techniques in the future could further enhance personalization and improve long-term effectiveness.

Integrated System Performance

The overall system was tested as a complete pipeline under real-time conditions. The system successfully performed:

- Face detection
- Attribute analysis
- Engagement estimation
- Advertisement display

within a short time frame.

The system maintained smooth real-time responsiveness with minimal delay between input and output. However, performance degradation was observed in scenarios involving multiple users or challenging environmental conditions such as poor lighting. These factors increased computational load and introduced additional complexity in processing.

Discussion

The integration of multiple modules into a unified pipeline was successful and demonstrated the feasibility of real-time intelligent retail analytics. The modular architecture ensured efficient data flow and allowed each component to contribute effectively to the overall functionality. This design also supports scalability, enabling future enhancements without significant restructuring.

3.2 Research Findings

Based on the comprehensive evaluation of the system, several key findings were identified that highlight both the strengths and practical implications of the proposed approach:

Effectiveness of Real-Time Computer Vision in Retail Environments:

The results clearly demonstrate that real-time computer vision can be effectively utilized for continuous customer analysis in retail settings. The system can capture live video streams and process them instantly to extract meaningful information such as facial attributes, emotional states, and behavioral patterns. This eliminates the need for manual observation or customer input, thereby reducing human effort and operational costs.

Furthermore, the ability to perform analysis in real time ensures that insights are immediately actionable. For example, advertisements can be dynamically updated based on the current audience rather than relying on pre-defined schedules. This significantly enhances the responsiveness of retail systems.

In addition, the system proves that modern computer vision techniques are mature enough to be deployed outside controlled laboratory conditions. Even though minor performance variations exist, the system maintains stable operation under practical conditions, confirming its suitability for real-world implementation.

Enhanced Advertisement Relevance through Multi-Dimensional Data Integration:

One of the most important findings is that integrating multiple data dimensions, demographic, emotional, and behavioral—substantially improves the quality and relevance of advertisement recommendations. Traditional advertising systems typically rely only on demographic segmentation, which provides a limited understanding of the customer. In contrast, system incorporates emotional context and engagement level, enabling a more comprehensive interpretation of user intent.

This multi-dimensional approach allows the system to adapt advertisements not only based on who the user is but also based on their current mood and level of interest. For instance, a highly engaged user displaying positive emotions may respond better to promotional or entertainment content, whereas a disengaged or neutral user may require different strategies.

As a result, the system achieves a higher level of personalization, which is critical in modern retail environments where customer attention is limited. This finding highlights the importance of combining multiple analytical perspectives to improve decision-making accuracy.

Importance of Engagement Analysis in Identifying High-Value Users:

The evaluation emphasizes that engagement analysis is a key factor in determining the effectiveness of advertisement delivery. By analyzing behavioral indicators such as how long a user remains in front of the display, whether they are looking directly at it, and how consistently they are detected across frames, the system can accurately estimate user engagement levels.

This capability allows the system to differentiate between users who are actively interacting with the display and those who are simply passing by. Such differentiation is crucial because it prevents unnecessary resource usage on low-interest users and focuses computational and advertising efforts on high-potential individuals.

Additionally, engagement analysis introduces a temporal dimension to the system, meaning that decisions are based on patterns over time rather than single-frame observations. This improves reliability and reduces noise in predictions. Overall, this finding highlights that behavioral analysis significantly enhances the practical value of intelligent retail systems.

Practicality and Effectiveness of Rule-Based Recommendation Systems:

The study found that a rule-based recommendation system can effectively translate analytical outputs into meaningful and context-aware advertisement decisions. By using predefined logical rules that combine age, gender, emotion, and engagement, the system can generate recommendations that align with expected marketing strategies.

One of the main advantages of this approach is its transparency. Each recommendation can be easily explained and traced back to specific input conditions, which is important for system validation and stakeholder trust. Additionally, rule-based systems are computationally efficient, making them suitable for real-time applications where speed is critical.

However, while the rule-based approach performs well in controlled scenarios, it lacks adaptability to changing user behavior over time. This suggests that future improvements could involve integrating machine learning-based recommendation techniques that learn from user interactions. Nevertheless, the current approach provides a solid and reliable foundation for initial system deployment.

Impact of Environmental Conditions on System Performance:

The evaluation reveals that environmental factors play a significant role in influencing system performance. Lighting conditions have a direct impact on the quality of facial feature extraction, which in turn affects the accuracy of age, gender, and emotion detection. Poor lighting or strong shadows can reduce model reliability.

Similarly, occlusions such as masks, glasses, or partial face visibility introduce challenges in accurate detection and classification. The presence of multiple users in the frame further increases

computational complexity, as the system must simultaneously track and analyze several individuals. This can lead to slight delays or reduced processing efficiency.

These findings highlight the importance of designing robust models that can handle variability in real-world environments. They also suggest the need for optimization techniques, such as improved preprocessing, better hardware support, or more advanced algorithms, to maintain consistent performance under challenging conditions.

Feasibility of Real-Time Integrated Intelligent Systems:

The overall results confirm that integrating multiple machine learning modules into a single real-time system is both feasible and effective. The system successfully combines face detection, attribute analysis, engagement estimation, and advertisement recommendation into a unified pipeline that operates continuously and efficiently.

The modular architecture of the system allows each component to function independently while still contributing to the overall objective. This not only improves system reliability but also makes it easier to upgrade or replace individual modules in the future.

Additionally, the system demonstrates that real-time processing can be achieved without significant delays, even when handling multiple analytical tasks simultaneously. This validates the concept of using integrated AI-driven solutions for smart retail environments and highlights their potential for large-scale deployment.

3.3 Discussion

The results demonstrate that the system successfully achieves its primary objective of delivering personalized advertisement recommendations based on real-time user analysis. By continuously capturing and processing live video input, the system can identify key user attributes such as age, gender, emotional state, and engagement level, and use this information to generate context-aware advertising content. This real-time capability ensures that the system responds dynamically to changes in the user environment, making it significantly more effective than traditional static advertising approaches.

The integration of multiple machine learning models plays a crucial role in enhancing the overall depth and quality of analysis. Instead of relying solely on demographic information, the system combines visual, emotional, and behavioral insights to form a more comprehensive understanding of each user. This multi-layered analysis allows the system to move beyond basic segmentation and deliver advertisements that are better aligned with user preferences and situational context. In particular, the inclusion of engagement analysis adds substantial value by enabling the system to identify users who are actively interacting with the display. By prioritizing these high-engagement users, the system improves the likelihood of successful advertising delivery and increases overall effectiveness.

However, the evaluation also highlights several limitations that affect system performance in real-world scenarios. Variations in environmental conditions, such as inconsistent lighting, shadows, and partial occlusions, can reduce the accuracy of computer vision models and impact the reliability of predictions. Additionally, the current rule-based recommendation system, while effective and easy to interpret, lacks the ability to adapt to evolving user behavior over time. This limitation restricts the system's ability to learn from past interactions and continuously improves its recommendations.

Despite these challenges, the system provides a strong and scalable foundation for intelligent retail analytics. The successful integration of artificial intelligence and computer vision technologies demonstrates the practical feasibility of deploying such systems in real-world environments. Furthermore, the modular design and promising performance outcomes indicate significant potential for future enhancements, such as incorporating adaptive learning mechanisms and improving robustness under varying environmental conditions. Overall, the results confirm that system is a viable and impactful solution for next-generation smart retail systems.

3.4 Limitations of the Study

Although the system demonstrates promising results, several limitations were identified during the evaluation phase that highlight areas for further improvement. One of the primary challenges observed is the sensitivity of the system to environmental conditions, particularly lighting variations. Under poor or uneven lighting, the quality of captured facial features is reduced, which negatively impacts the accuracy of face detection as well as subsequent tasks such as age, gender, and emotion classification. This limitation indicates that the system's performance is highly dependent on the quality of data input.

In addition to lighting issues, the system also struggles with extreme facial angles and partial occlusions. When users are not directly facing the camera or when parts of the face are covered by objects such as masks, glasses, or hair, the model's ability to extract reliable features is significantly reduced. This can lead to incorrect predictions or missed detections, particularly in real-world retail environments where user movement is unpredictable.

Emotion detection presents another challenge due to the complexity and subjectivity of human facial expressions. Subtle or ambiguous expressions often share overlapping visual characteristics, making it difficult for the model to distinguish between certain emotional states. As a result, the accuracy of emotion classification decreases in such scenarios, leading to occasional inconsistencies in prediction outputs.

Furthermore, the current recommendation system is based on predefined rules, which, while effective for initial implementation, lacks adaptability. The system does not learn from user interactions or feedback over time, limiting its ability to refine and optimize recommendations

based on evolving user behavior. This restricts the system's potential for long-term personalization and continuous improvement.

Another important limit arises when multiple users are present within the camera frame. In such situations, the system must simultaneously detect, track, and analyze several individuals, which increases computational complexity. This can lead to a reduction in processing speed and may affect real-time responsiveness, especially when hardware resources are limited.

Overall, these limitations highlight the challenges associated with deploying real-time computer vision systems in dynamic, real-world environments. However, they also provide valuable insights into potential areas for enhancement, such as improving model robustness, optimizing system performance, and incorporating adaptive learning techniques to strengthen the system's capabilities.

3.5 Future Work

Based on the limitations identified during the evaluation, several directions for future enhancement can be proposed to improve the performance, robustness, and scalability of the system. One of the primary areas for improvement is the system's sensitivity to environmental conditions, particularly lighting variations. Future work can focus on incorporating advanced image preprocessing techniques such as adaptive histogram equalization or low-light image enhancement models to improve input quality under poor lighting conditions. Additionally, the use of infrared or depth-based sensors could be explored to reduce dependency on visible light conditions.

Another important improvement involves handling extreme facial angles and partial occlusions more effectively. Future versions of the system could integrate more robust deep learning architectures or multi-angle face detection models that are trained on diverse datasets containing varied poses and occluded faces. This would significantly enhance detection reliability in real-world retail environments where user movement is unpredictable.

Emotion detection accuracy can also be improved by adopting more advanced models such as transformer-based architectures or temporal deep learning approaches that consider sequences of facial expressions rather than single-frame analysis. Incorporating contextual and temporal information would help reduce ambiguity in emotion classification and improve consistency in predictions.

In terms of recommendation systems, a key enhancement would be the transition from a rule-based approach to an adaptive machine learning-based recommendation engine. By incorporating reinforcement learning or collaborative filtering techniques, the system could learn from user interactions over time and continuously refine its advertisement strategies. This would enable more personalized and dynamic recommendations that evolve with user behavior.

Additionally, system performance under multiple-user scenarios can be improved through better optimization strategies, such as parallel processing, model compression, or the use of lightweight neural network architectures. Edge computing techniques could also be considered to distribute computational load and maintain real-time responsiveness even in high-traffic environments.

Overall, future work should focus on improving robustness, adaptability, and scalability of the system. These enhancements would further strengthen the practicality of system and support its deployment in large-scale intelligent retail environments.

3.6 Summary of Each Student's contribution

The successful completion of this research project was the result of collaborative efforts by all team members. Each member contributed to different aspects of the system, including model development, system integration, frontend and backend implementation, and testing. The integration of these individual efforts resulted in the successful development of a complete real-time retail analytics system. The following outlines each member's individual contribution:

Student 1: Fonseka W.S.M (IT22109712)

Component: Age and gender detection module.

The contributions include:

- Designing and implementing the convolutional neural network (CNN) for age and gender classification
- Preparing and preprocessing datasets for model training
- Training, tuning, and evaluating the model
- Optimizing the model for real-time performance
- Integrating the model into the overall system pipeline

Student 2: Samaraweera W.D.U.I (IT22258526)

Component: Emotion detection module of the system.

Key contributions include:

- Designing and implementing the emotion recognition model
- Training the model using facial expression datasets
- Evaluating model performance and improving prediction stability

- Implementing temporal smoothing techniques for real-time prediction
- Integrating the emotion detection module with the preprocessing pipeline

Student 3: Abeykoon S.N (IT22184030)

Component: Engagement analysis and advertisement recommendation system.

The contributions include:

- Designing the engagement analysis mechanism based on behavioral indicators
- Identifying and extracting key features such as attention duration, head orientation, and presence consistency
- Developing a scoring model to quantify user engagement levels
- Implementing temporal aggregation techniques to stabilize engagement predictions
- Integrating the engagement module with the preprocessing pipeline

Student 4: Rajana H.T.R(IT22071934)

Component: Reaccommodation system

Key responsibilities include:

- Designing and implementing the rule-based advertisement recommendation system
- Mapping user profiles (demographic, emotional, and engagement data) to advertisement categories
- Managing advertisement selection and real-time display logic
- Testing and refining recommendation outputs for accuracy and relevance
- Supporting system integration and ensuring smooth data flow between modules

CONCLUSION

This study presented the design and development of system, an intelligent retail analytics system that leverages computer vision and machine learning techniques to analyze customer attributes and behavior in real time. The system integrates multiple functional components, including age and gender detection, emotion recognition, engagement analysis, and advertisement recommendation, into a unified and automated framework. The primary aim of the study was to address the limitations of traditional retail environments, which often lack personalization and rely on static advertising methods. By incorporating real-time data processing and intelligent decision-making, the system demonstrates how artificial intelligence can be effectively applied to enhance customer engagement and improve the overall retail experience.

The objectives defined at the beginning of the study were successfully achieved through the implementation and evaluation of the proposed system. The age and gender detection module was effectively developed and integrated into the system, providing reliable demographic classification. The emotion recognition component contributed meaningful contextual insights by identifying user emotional states, while the engagement analysis module successfully estimated user interaction levels based on behavioral patterns. Furthermore, the advertisement recommendation system was able to generate personalized outputs by combining results from all analytical modules. The integrated system also demonstrated the ability to operate in real time under practical conditions, confirming that the overall objectives of the study were met.

This research makes several significant contributions to the field of intelligent retail systems. It presents the development of a real-time retail analytics framework that combines computer vision and machine learning techniques for customer analysis. The study also introduces an integrated approach that combines demographic, emotional, and behavioral data within a single system, enabling a more comprehensive understanding of user behavior. In addition, it demonstrates a practical implementation of a model-assisted advertisement recommendation system that operates in real time. Overall, the study highlights the feasibility and effectiveness of deploying intelligent systems to enhance customer engagement and support data-driven marketing strategies in retail environments.

Although the system achieved its intended goals, several limitations were identified during the evaluation process. The performance of the system tends to decrease under poor or inconsistent lighting conditions, which affects the accuracy of facial analysis. Additionally, the system faces challenges when dealing with extreme facial angles and partial occlusions, which can reduce detection reliability. In scenarios involving multiple users, the system experiences increased computational complexity, which may impact real-time performance. Furthermore, the current recommendation mechanism is based on a rule-based approach, which limits its ability to adapt dynamically to changing user behavior over time.

Future improvements to the system should focus on enhancing both performance and adaptability. The integration of advanced machine learning techniques, such as reinforcement learning or deep

learning-based recommendation models, could enable more dynamic and personalized advertisement generation. Expanding and diversifying training datasets would further improve model accuracy and generalization across different environments. Additionally, the use of cloud-based or edge computing solutions could support large-scale deployment and improve system efficiency. Incorporating additional behavioral and contextual features could also enhance the accuracy of engagement analysis, while further optimization is required to ensure stable performance in complex real-world scenarios.

In conclusion, the system successfully demonstrates the potential of combining artificial intelligence and computer vision to transform traditional retail environments into intelligent and interactive spaces. By enabling real-time customer analysis and personalized advertisement delivery, the system effectively bridges the gap between physical and digital retail experiences. The findings of this study highlight the significant role of intelligent systems in improving customer engagement, enhancing marketing effectiveness, and supporting data-driven decision-making. With further refinement and development, such systems have strong potential to become a key component of future smart retail technologies.

GLOSSARY

Accuracy

A performance metric used to evaluate machine learning models, representing the proportion of correct predictions out of total predictions.

Artificial Intelligence (AI)

A field of computer science focuses on creating systems capable of performing tasks that typically require human intelligence.

Camera Input

The live video stream captured from a camera device, used as the primary data source for the system.

CNN (Convolutional Neural Network)

A type of deep learning model commonly used for image processing tasks such as classification and detection.

Computer Vision

A field of artificial intelligence that enables machines to interpret and analyze visual data from images or videos.

Deep Learning

A subset of machine learning that uses neural networks with multiple layers to learn complex patterns from data.

Edge Computing

A computing paradigm where data processing is performed close to the data source, reducing latency and improving real-time performance.

Engagement Analysis

The process of evaluating user interaction and attention levels based on observable behavior such as presence duration and head orientation.

Face Detection

The process of identifying and locating human faces within images or video frames.

Facial Landmarks

Key points on a face (such as eyes, nose, and mouth) used for alignment and feature extraction.

FastAPI

A modern web framework used for building high-performance APIs in Python.

Inference

The process of using a trained machine learning model to make predictions on new data.

Latency

The time delay between input (video capture) and output (system response).

Machine Learning (ML)

A subset of artificial intelligence that enables systems to learn from data and make predictions without explicit programming.

MongoDB

A NoSQL database used for storing unstructured or semi-structured data.

Neural Network

A computational model inspired by the human brain, used for pattern recognition and data analysis.

ONNX Runtime

A high-performance inference engine used to execute ONNX models efficiently.

OpenCV (Open-Source Computer Vision Library)

A library used for real-time computer vision tasks such as image processing and video analysis.

Preprocessing

The process of transforming raw data into a suitable format for machine learning models.

Precision

A metric used to evaluate classification performance, representing the proportion of true positive predictions among all positive predictions.

Real-Time Processing

The ability of a system to process data and produce outputs instantly or with minimal delay.

Recommendation System

A system that suggests relevant content or items based on user data and behavior.

Scalability

The ability of a system to handle increased workload or expand to larger environments.

Softmax Function

A mathematical function used in classification models to convert outputs into probability distributions.

TensorFlow

An open-source machine learning framework used for building and training deep learning models.

Webcam

A camera device used to capture live video input for the system

REFERENCES

- [1] M. &. Company, "The future of personalization in retail," 2021.
- [2] Gartner, "AI in retail analytics report," 2022.
- [3] R. Szeliski, *Computer Vision: Algorithms and Applications*, Springer, 2010.
- [4] e. a. Ian J. Goodfellow, "Challenges in representation learning: A report on three machine learning contests," *Neural Networks*, 2015.
- [5] W. D. Shan Li, "Deep Facial Expression Recognition: A Survey," *IEEE Transactions on Affective Computing*, 2022.
- [6] P. E. a. W. V. Friesen, "Constants across cultures in the face and emotion," *Journal of Personality and Social Psychology*, 1971.
- [7] D. C. M. H. M. Ali Mollahosseini, "Going deeper in facial expression recognition using deep neural networks," 2016.
- [8] H. L. a. Y. C. M. Al-Hubaishi, "A federated deep learning approach for distributed denial-of-service attack detection," *Computers & Security*, 2023.
- [9] C. Z. L. C. Y. L. Cheng Chang, "Body Orientation and Dwell Time Fusion for Audience Engagement Detection," *ICMI-MLMI*, 02 October 2018.
- [10] S. R. a. P. Norvig, *Artificial Intelligence: A Modern Approach*, 3rd ed., Pearson, 2010.
- [11] Deloitte, "Retail trends and customer experience transformation," 2022.
- [12] K. H. R. G. a. J. S. S. Ren, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2017.
- [13] Y. L. e. al., "Robust face detection in unconstrained environments," *IEEE Transactions on Image Processing*, 2019.
- [14] X. A. a. J. Basilico, "Recommender systems in industry: A Netflix case study," in *IEEE Computer*, 2012.
- [15] Y. B. a. G. H. Y. LeCun, "Deep learning," *nature*, 2015.
- [16] K. K. S. a. A. A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *ICLR*, 2015.

- [17] S. E. C. e. al, "Age estimation using a hierarchical classifier based on global and local facial features," in *IEEE FG*, 2011.
- [18] M. Z. U. e. al, "Facial expression recognition using deep CNN," *IEEE Access*, 2019.
- [19] B. F. a. J. Luetttin, "Automatic facial expression analysis: A survey," *Pattern Recognition*, 2003.
- [20] R. G. S. e. al., "Human attention detection using head pose estimation," in *IEEE ICIP*, 2020.
- [21] G. A. a. A. Tuzhilin, "Toward the next generation of recommender systems," *IEEE Transactions on Knowledge and Data Engineering*, 2005.
- [22] W. L. e. al., "SSD: Single shot multibox detector," in *ECCV*, 2016.